

12.319

AI25BTECH11033 - Spoorthi N

Question:

Let M be the real vector space of 2×3 matrices with real entries. Let $T : M \rightarrow M$ be defined by

$$T\left(\begin{bmatrix} x_1 & x_2 & x_3 \\ x_4 & x_5 & x_6 \end{bmatrix}\right) = \begin{bmatrix} -x_6 & x_4 & x_1 \\ x_3 & x_5 & x_2 \end{bmatrix}. \quad (1)$$

The determinant of T is _____

(MA 2013)

Solution:

We are given the transformation

$$T\left(\begin{pmatrix} x & y & z \\ u & v & w \end{pmatrix}\right) = \begin{pmatrix} -w & u & x \\ z & v & y \end{pmatrix}. \quad (2)$$

Represent input as a vector;

$$\begin{pmatrix} x & y & z \\ u & v & w \end{pmatrix} \longleftrightarrow \begin{bmatrix} x \\ y \\ z \\ u \\ v \\ w \end{bmatrix}. \quad (3)$$

Transformation on the vector;

$$T\left(\begin{pmatrix} x \\ y \\ z \\ u \\ v \\ w \end{pmatrix}\right) = \begin{pmatrix} -w \\ u \\ x \\ z \\ v \\ y \end{pmatrix}. \quad (4)$$

Matrix of transformation;

$$A = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}. \quad (5)$$

Determinant:

$$\det(A) = (-1) \times (-1) = 1. \quad (6)$$

$$\det(T) = 1 \quad (7)$$